

| Problem

Design an AI based solution for the field of pharmaceuticals where they face a critical challenge in data-driven decision making due to discrepancy in EHRs from diverse sources from clinical records to genomic information. Which in turn hinder the timely and accurate analysis of patient outcomes, treatment effectiveness, and disease progression.

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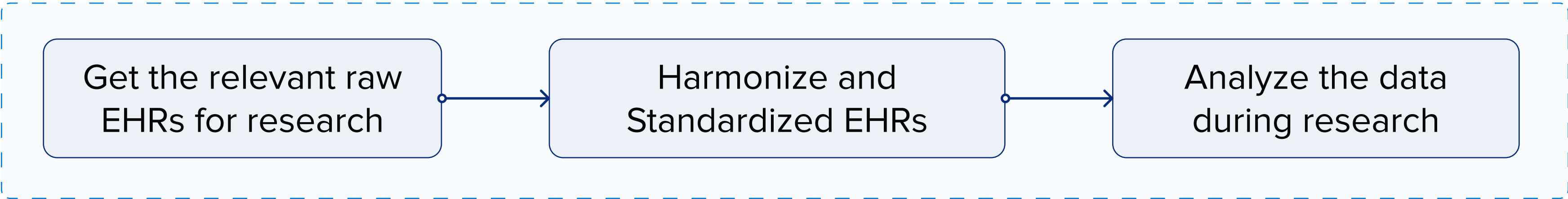
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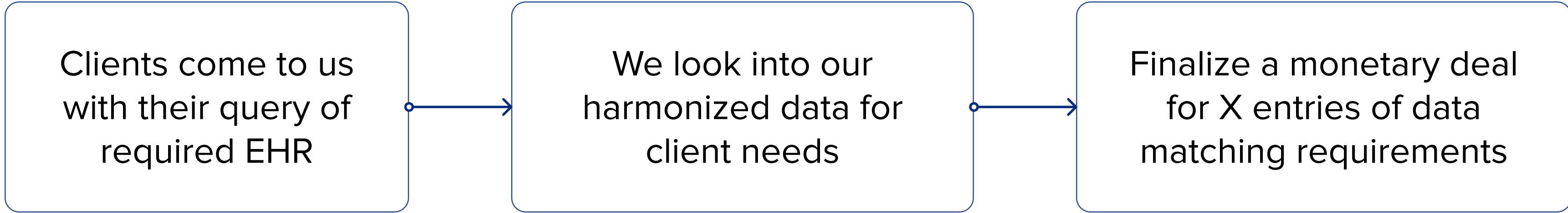
Problem	Stakeholders	Solution	Pricing	Metrics & Pitfalls
Understanding the current process <pre> graph LR A[Need to create a drug] --> B[In-depth data required for research] B --> C[Drug Discovery Phase] C --> D[Pre Clinical Research] D --> E[Clinical Trial Phase] E --> F[Post Trial Data Analysis] F --> G[FDA Approval for drug] G --> H[Drug Production starts] </pre> Pharma companies require a lot of real world sample data (EHR) for disease biology research, drug discovery and identifying suited people for clinical trial. However every institution (AMCs, Insurance claims, Patient Registries, Clinical Trial and Genomic Databases) has a different EHR format, which causes extreme inconsistencies when combined hence affects the analysis, which ultimately results into errors (eg. mismatched data fields, duplicate records, date/time errors and blank fields.) Reasons for inconsistency in EHR Heterogeneous Data Sources: Healthcare data comes from various sources, each source has it's own method for recording and utilising information. Lack of Standardization: There is no universal standard for EHR, which results in different formats and terminologies being used across different platforms. Manual Data Entry: Human error in manual data entry is responsible for mistakes occurring in the form of typos, omissions, or misinterpretations. Temporal Discrepancies: Changes in patient conditions over time and delays in updating records can lead to outdated information being present in the EHRs.			60% EHR data influence formulary decisions for approximately 60% of new drugs 94% 94% of hospital CIOs reported EHR interoperability as a major hurdle. 25% Error rates in USA in EHRs ranging from 2.3% to 26.9% across various studies. 20% Data quality issues could lead to a 20% increase in clinical trial durations. Potential Impact of solving it - Enhancing data accuracy could reduce trial times, potentially saving up to \$600,000 daily for late-stage trials (Source: PRMA) - Clearer data improves approval rates, with every 1% increase in approval likelihood equating to an estimated \$50 million in revenue for a new drug (Source: McKinsey) Improved data quality could increase operational efficiency by 30%, reducing the time to insight for critical decision-making (Source: Deloitte). A 10% improvement in data management could save up to \$200 million from the R&D budget (Source: PRMA) - Improved data could expedite patient access to new treatments by 3-6 months, benefiting patient health and company revenues (Source: IMS Health). Assumptions Data Availability: We have access to EHR data, despite its fragmented nature, is largely obtainable through partnerships or existing public database & APIs. Quality of Source Data: The source EHR data, while imperfect, contains sufficient detail to be enhanced through AI-driven curation processes. Compliance Feasibility : Regulatory bodies like HIPAA will support the use and interoperability of EHR in drug development, with a focus on data quality and patient privacy using data masking & de-identification.	

Introducing M-Intel



MIntel provides service to both pharma companies who don't have the raw EHRs for research to begin with and companies who have EHRs from multiple source but the data can not be merged because it is not harmonized or present in a standard format. In addition to this, it also equipped with semantic search model to help researchers to identify patterns in the process of developing a new drug.

1. Providing relevant harmonized data for research



This will majorly target medium sized pharma companies because they often have limited resources compared to large companies and by out-sourcing data acquisition and harmonization through a same service will save cost and complexity of maintaining a large scale data harmonizaion operation

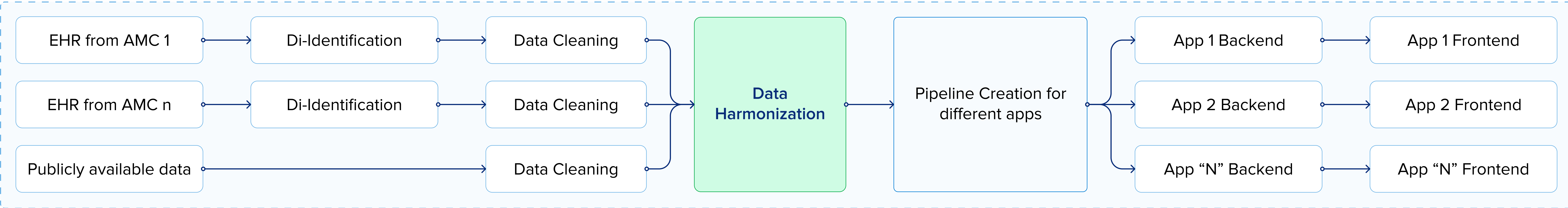
De-identification: We can use Named Entity Recognition (NER) models, like BERT to identify and redact personal identifiers from datasets to prevent identification of individuals. eg system recognizes the patient's name & DOB, and replaces them with anonymous identifiers like name with "Patient 12345" and change DOB to just the year.

Data Cleaning: Learning model like DBSCAN can be trained to identify anomalies or inaccuracies correcting them on basis of other attributes to improve quality. eg. a weight entry is listed as 6500 kgs, system recognizes this as an obvious error, and corrects it on basis of age, height etc and also flags it system generated value.

Data Harmonization and Normalization: Semantic AI can be used to understand the context and meaning behind different data representations and map them to a standardized format. eg. One hospital records blood pressure as "BP" while another records it as "Blood Pressure" then our system would detect this and map it to our decided schema

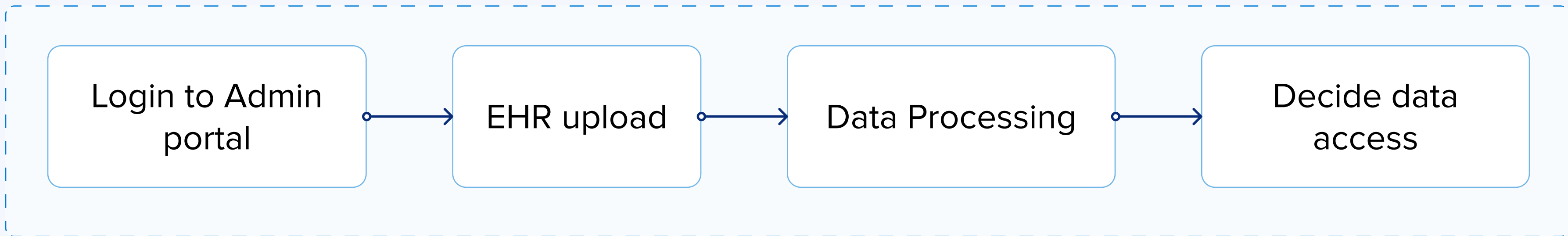
Pipeline Creation: Our overall database would have millions of entries and navigating through them is a time taking process, in pipeline creation a separate schema will be made as per the client requirement, which we save time as well as resources.

Various app backend and frontend: Backend with access to filtered data by the pipeline will be the server-side of the client's app dealing with data management, role management, permission handling and processing which will power user interface and experience components of an app, through which users interact.



2. Ingesting their own EHR data to be harmonized

It allows pharma companies to **upload their own EHR data (encrypted at both end), with option of data processing**. With minimal human input, it ensures privacy, accuracy, and standardized data ready for analysis, saving time and enhancing research quality. This feature is **ideal for large-sized pharma firms those who have their own data sources and do not trust other organization with it.**

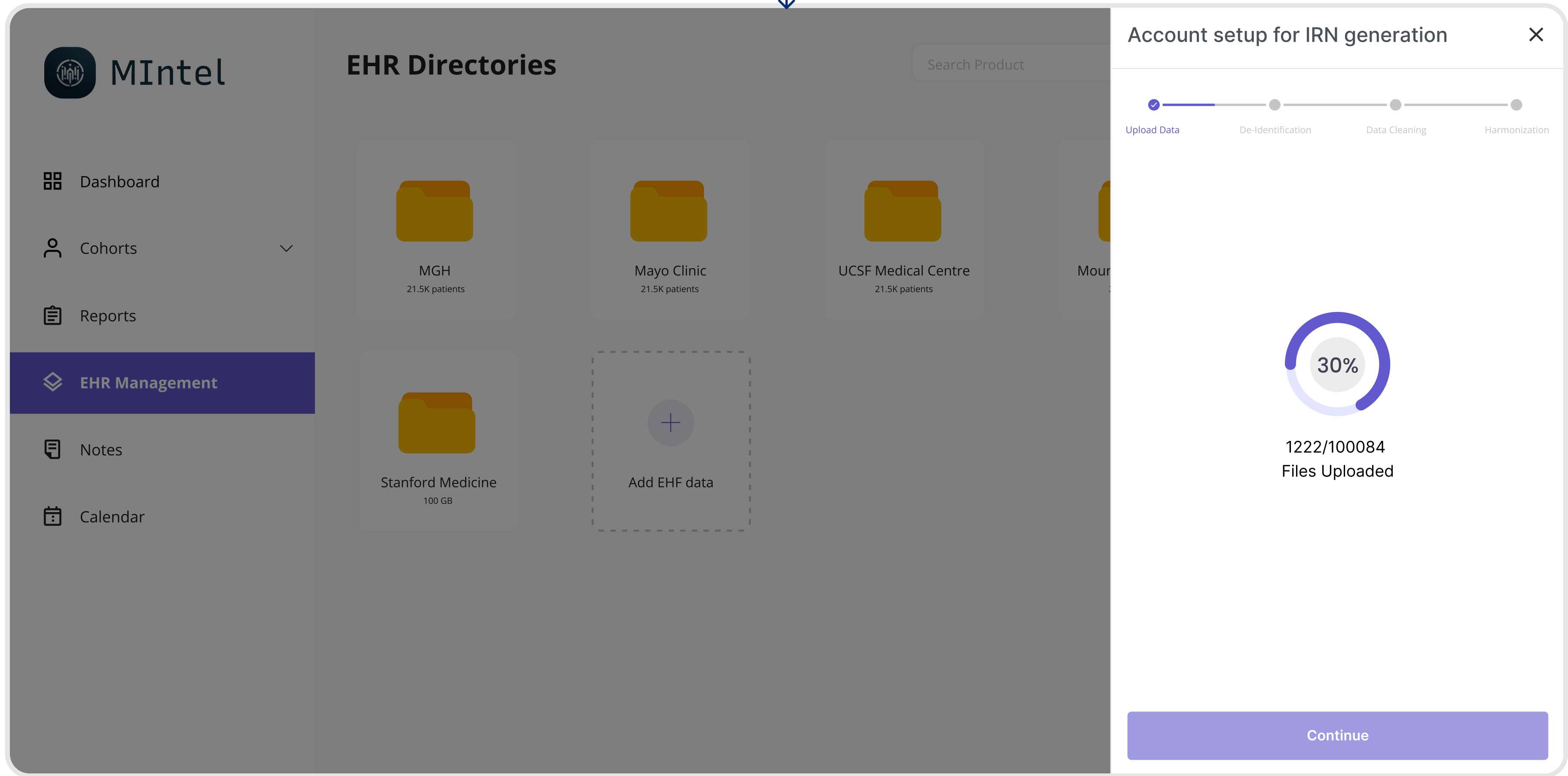
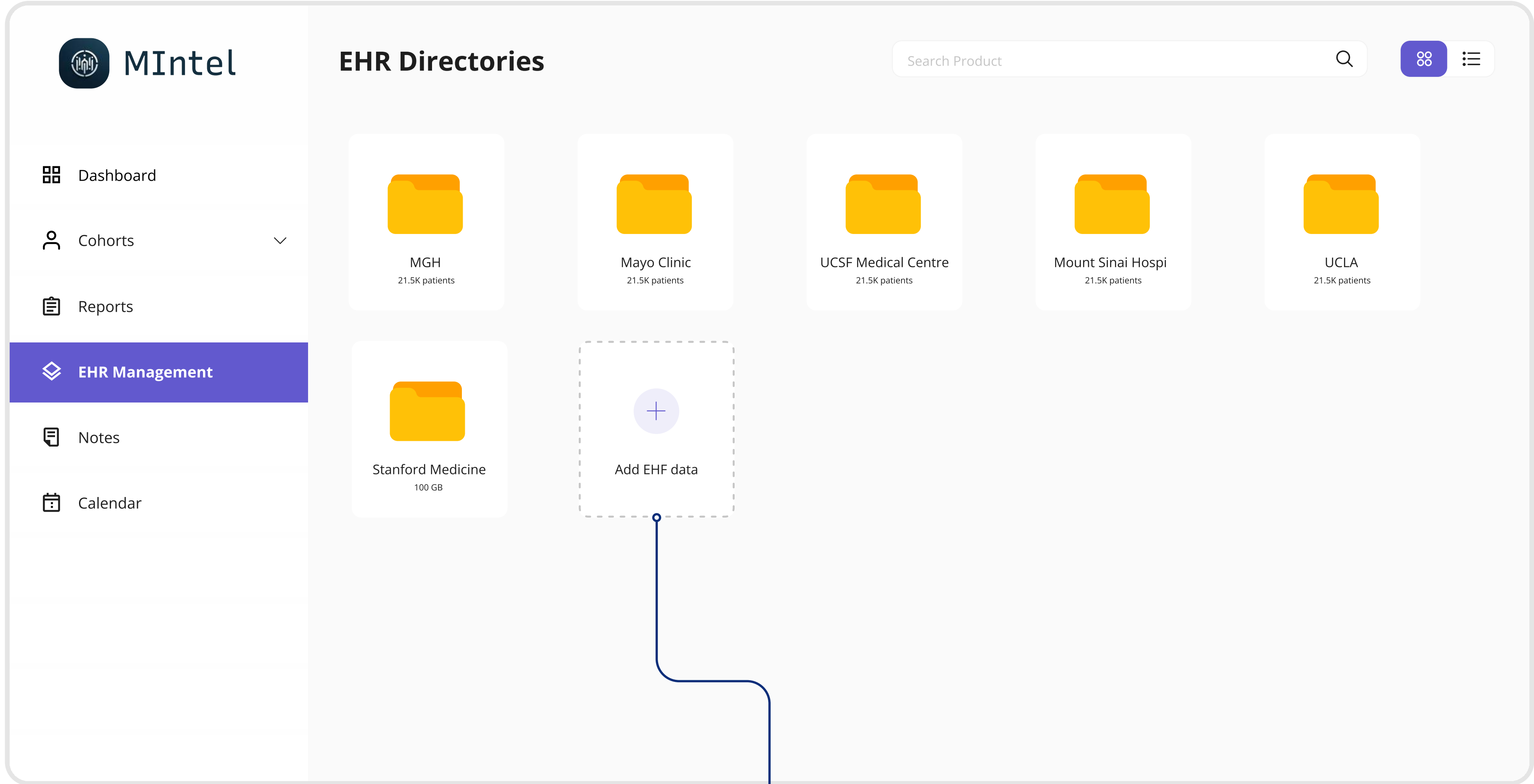


Login to Admin Portal: A multi-factor authorization portal designed for admins, where they would be able to manage EHRs, Reports, Cohorts and can also access project’s activities, including recent uploads, processing status, and access logs.

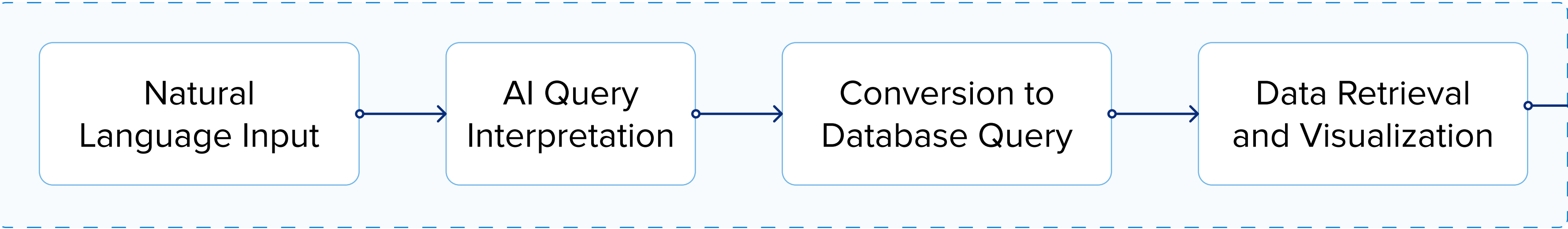
EHR Upload: Within the admin portal, there is a dedicated section for EHR data upload. Admins can upload files from their system or can connect with external APIs to fetch hosted data. The portal accepts various file formats and sizes, supporting the diverse types of EHR data that might be collected from different sources.

Data Processing: After the EHR data is uploaded, it enters the AI-driven processing stage. This involves several **automated steps including: De-Identification, Data Cleaning and Harmonization**. Users can track and intervene the progress of these processes through the admin portal and receive notifications upon completion.

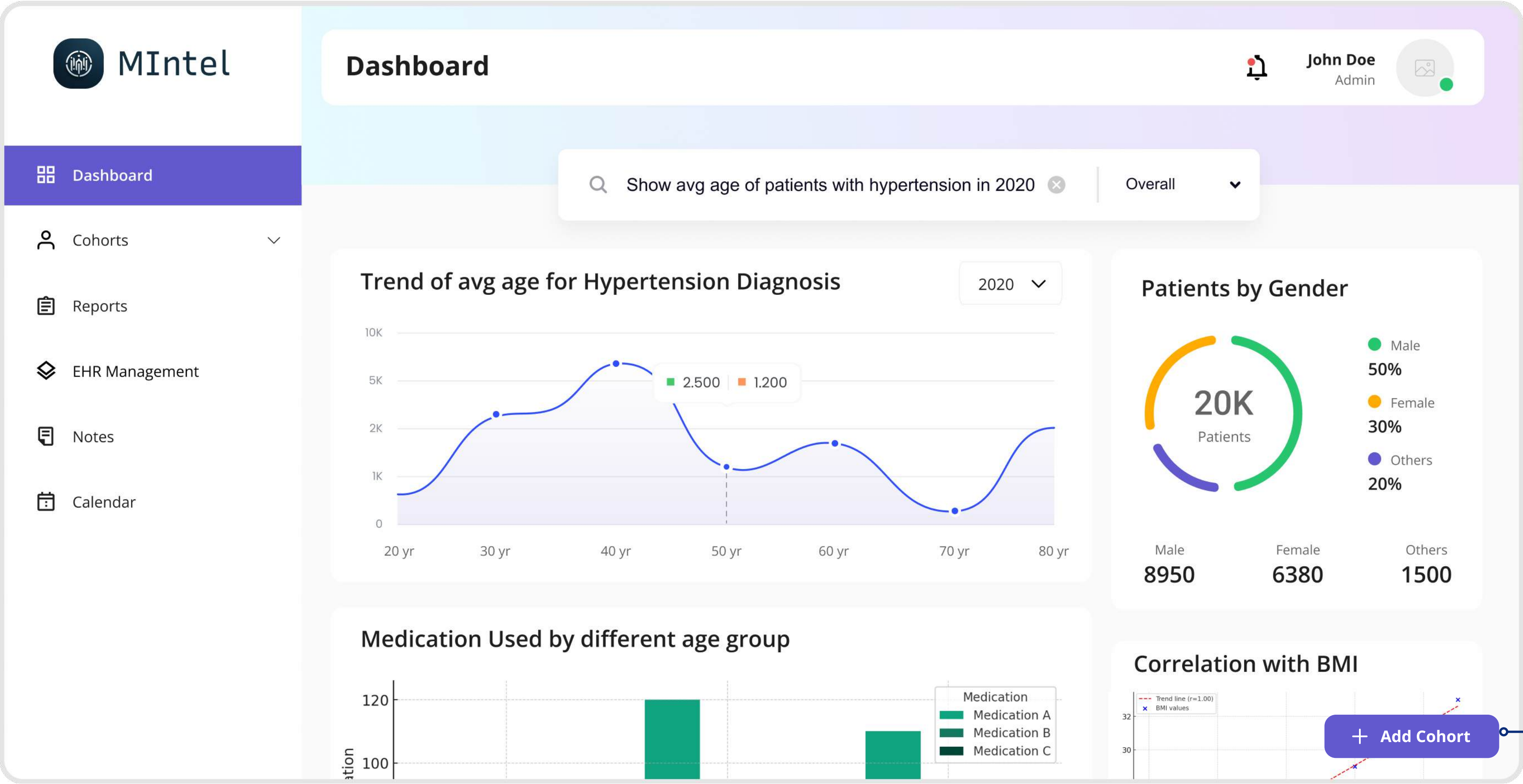
Decide Data Permissions: Once data processing is complete, users can set permissions for who within their organization can access the data on basis of roles.



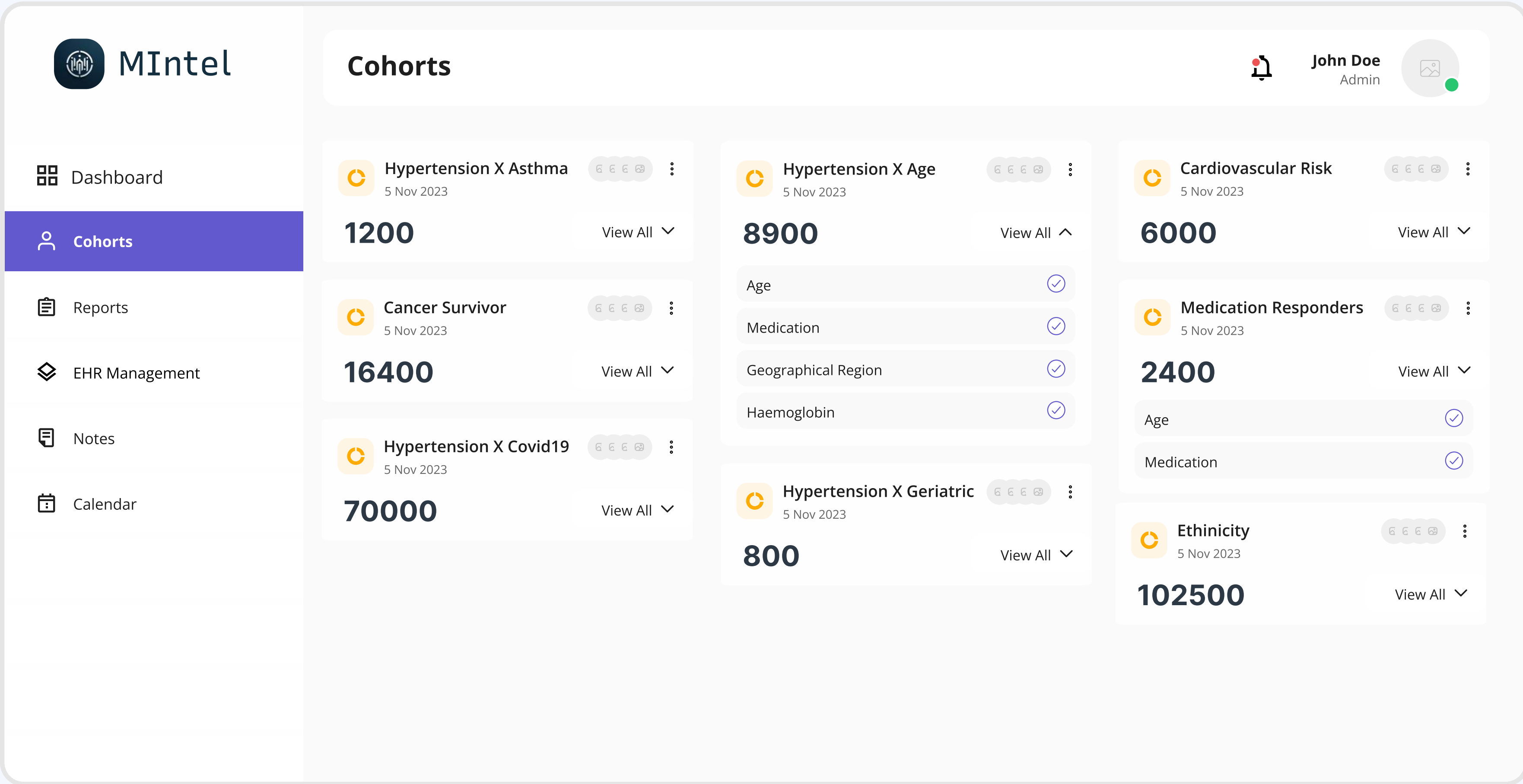
3. Smart search to convert natural questions to DB queries



This feature provides an intuitive way for users to interact with complex databases using simple English queries eg. “Show me the average age of patients with hypertension from the 2020 dataset.”, behind the scene search engine uses models like BERT or GPT-3.5 (trained on medical and pharmaceutical text) to understand the context and intent behind the user’s natural language and convert it into a SQL query, which in turn process (categorization) & fetch the data. The results are presented to the user in an easily digestible format, typically as graphs, charts, or tables eg. a bar graph showing the avg age of hypertension patients



4. Patient Cohort creation for research



It offers a streamlined process for users to define and analyze specific patient populations within their healthcare data. Users selects adds various attributes, such as age and diagnosis, to set up the criteria for their cohort. The system then dynamically generates a matching patient segment with their fine-tune parameters. Cohort creation allows for the saving and sharing of it, facilitating collaborative research while adhering to privacy standards. With AI-driven suggestions and real-time data processing, this feature simplifies complex data analysis, making it accessible and efficient for all user levels within a pharmaceutical organization.

	Problem	Stakeholders	Solution	Pricing	Metrics & Pitfalls
Pricing Strategy	As a part of pricing GTM, our primary focus is find a sweet point between recovering our initial costs and it should be affordable to since we would be entering a new market. According to the strategy, we need to focus on gaining market share and increasing our revenue through sales. <pre> graph TD TC[Total costs] --> FC[Fixed Costs] TC --> VC[Variable Costs] FC --> SL[Software Licenses] FC --> AMD[AI model development] FC --> EC[Employee cost] FC --> IC[Infrastructure cost] VC --> EO[EHR Outsourcing] VC --> SSC[Server (scalability) Cost] VC --> MVC[Monthly variable cost] VC --> MS[Marketing & Sales] </pre>				
Exploring	1. Price Skimming: The price is set for high-paying customers, and then lowered over a period of time. This gives higher return on investment. 2. Penetration pricing: It is just opposite of price skimming, where initially price is set lower and then increased gradually over time. 3. Value based pricing: This pricing is based on customers perceived value of the product rather than the actual cost of the product. 4. Competitive pricing: When there are a lot of competitors, it is better to set prices at a lower side to prevent customers to go to our competitors. 5. Cost plus pricing: In this type of pricing, the Product is priced on the basis of the cost. Simply some percentage is added over the cost.				
Value Based Pricing	Subscription-Based Model (Client using our generic EHR database) - Tiered Access: Different subscription tiers offering varying levels of access and features, such as the number of user accounts, volume of data storage, and processing power for data analysis. - Flexible Plans: Monthly and annual subscription options to provide flexibility and encourage long-term commitment. Enterprise Custom Contracts (Client with specific EHR requirements) - Bespoke Solutions: Custom pricing for large pharmaceutical companies requiring extensive use of the platform, including integration with existing systems, custom features, or dedicated support. - Volume Discounts: Reduced rates for high-volume data purchase or for companies that commit to a certain level of usage. Freemium Model (Requirement of only EHR Harmonization) - Basic Features for Free: Limited access to the platform for free, allowing users to upload data and run a certain number of queries or create a limited number of cohorts. - Premium Features: Advanced features, such as more complex queries, larger data uploads, or additional cohort analyses, available for a fee.				

	Problem	Stakeholders	Solution - I	Solution - II	Metrics & Pitfalls
Metrics to look	Acquisition	Activation	Retention	Engagement	Monetization
	• #number of customers acquired • Cost Per Acquisition	• User Onboarding Completion Rate • Successful First Query • Successful First Cohort Creation	• User Onboarding Completion Rate • Monthly Active users • Weekly Active users • Churn Rate	• Avg Session Duration • #Number Query per Session • #Number of Cohort Created per customer	• Average Revenue per User • Conversion (Free trial to paid customers)

Pitfalls and Mitigations

Potential Pitfalls	How to reduce risk?
• Mishandling sensitive data can lead to breaches in regulations like HIPAA.	• Implement robust encryption, access controls, and regular audits to de-identification techniques
• AI algorithms might be leading to the inaccurate analysis.	• Establish rigorous data cleaning and validation processes. Continuously monitor data quality and retrain AI
• Over-reliance on AI could lead to potentially overlooking errors.	• Educate users on the strengths and limitations of AI. Implement checks and balances for human verification
• Regulations governing EHR data are subject to change, leave the platform non-compliant.	• Engage with legal experts to anticipate and respond to regulatory shifts proactively.

What more can be done?

- Clinical Trial Matching Module:** We can develop an advanced system that uses AI to match patients with ongoing clinical trials based on their health data, potentially accelerating recruitment and improving trial diversity.
- Decentralized Data Ecosystems:** Blockchain can be utilized to create a decentralized data exchange where researchers can share data securely and patients can control who accesses their information.
- Global Regulatory Compliance:** Currently I have considered only US market, however we can extend the product in such a way that it would automatically adapt data handling and privacy measures to comply with global regulations, not just HIPAA.
- Patient Engagement Tools:** These modules will allow for direct patient engagement and data collection, such as digital consent forms, patient-reported outcomes and ata collection capabilities to include data from wearable devices and Internet of Things (IoT) healthcare applications.

PROBLEM

Projected into 2030, the growing technology will expose an individual's digital footprints publicly, hence raising the privacy concerns to its highest.

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What makes it a 2030 problem?

As we move towards 2030, our world will see a lot more data being created and shared due to the global nature of social media, an increase in internet penetration and advancement in AI and the adoption of IOT (e.g. smart home devices, wearables, etc) & VR devices.

However, this also means that our personal information will be more exposed, and the tools required to misuse it (e.g. deep fake, voice synthesiser) will be easier to access. Making privacy a big concern that would result in need for public awareness and initiatives from govt.

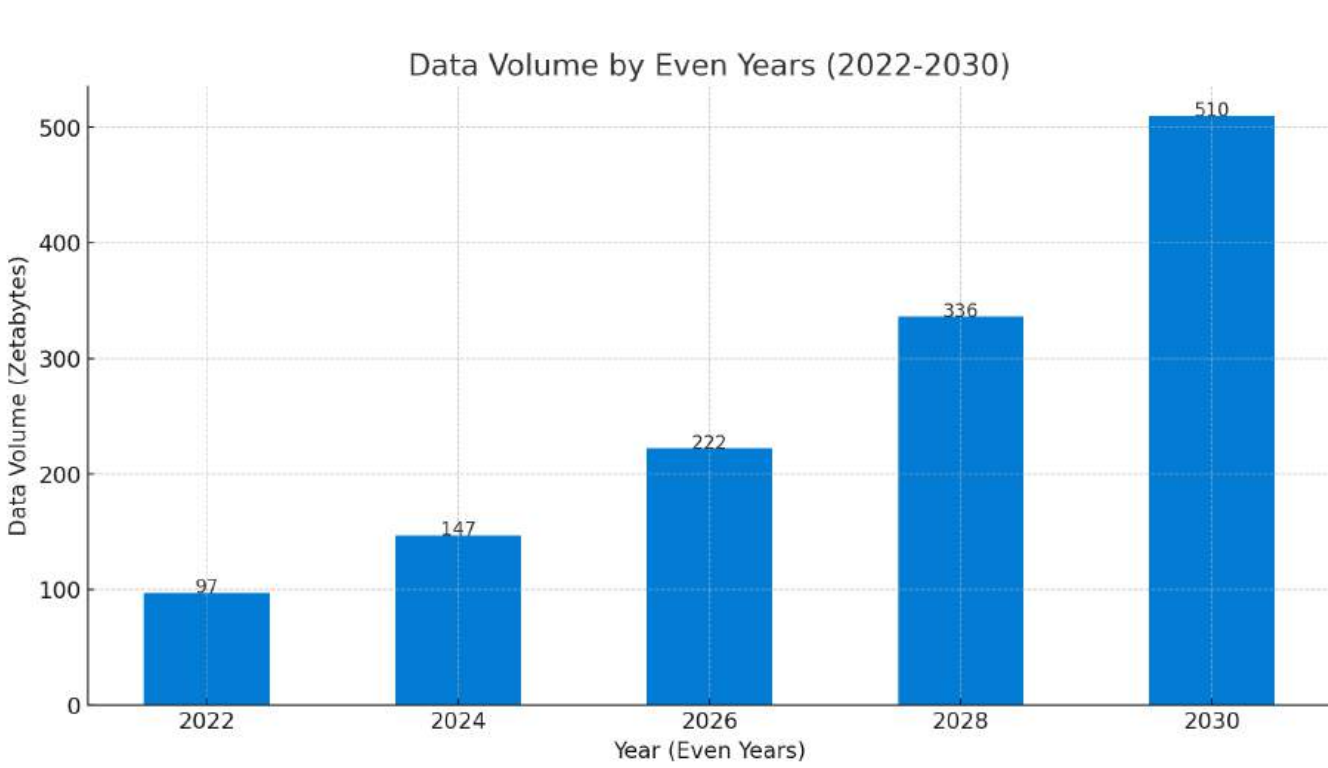
What are digital footprints?

- 👤 Personal Data (name, age, city etc.) | Source: Social Media, Documents
- 👁️ Biometric Data (face, retina, fingerprint etc.) | Source: Devices, Associated Organizations
- 🛒 Behavioural Data (preferences and habits) | Source: Social Media, Food, e-Comm apps etc
- 📄 Health & Financial Data (Records) | Source: Documents, health tracking/bank apps

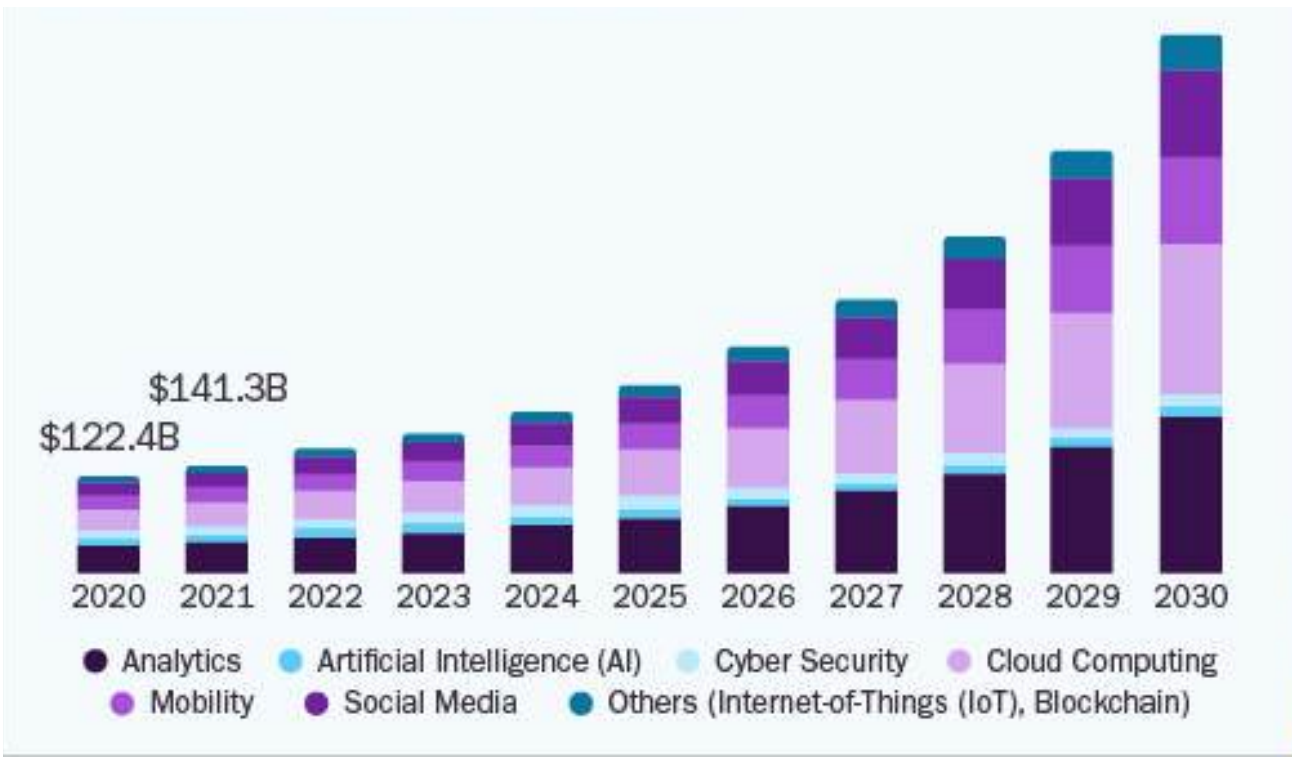
Problem Validation (Secondary Research*)

- Strait research suggests: The global big data security market size was worth USD 24 billion in 2021 and is projected to reach USD 115 billion by 2030, registering a CAGR of 19%.
- According to Varonis, only 5% of companies keep data encrypted at their end and two-thirds of companies have 1,000+ sensitive files open to every employee.
- By 2030, a complex mesh of 91 billion devices worldwide is expected, equating to over 10 connected devices per person (IOT device market will skyrocket at \$3 trillion in 2030), where every connected device potentially expands the surface for privacy breaches.

Note*: Primary research showed that people are worried about their privacy but aren't taking any measures



Global data is projected to exceed 510 zettabytes by 2030 (500% growth in 6-7 years with 23% CAGR)



The digitalization market of US, valued at USD 731.13 billion in 2022, is forecasted to at a CAGR of 26.7%

Potential impacts of solving it

Impact on Businesses

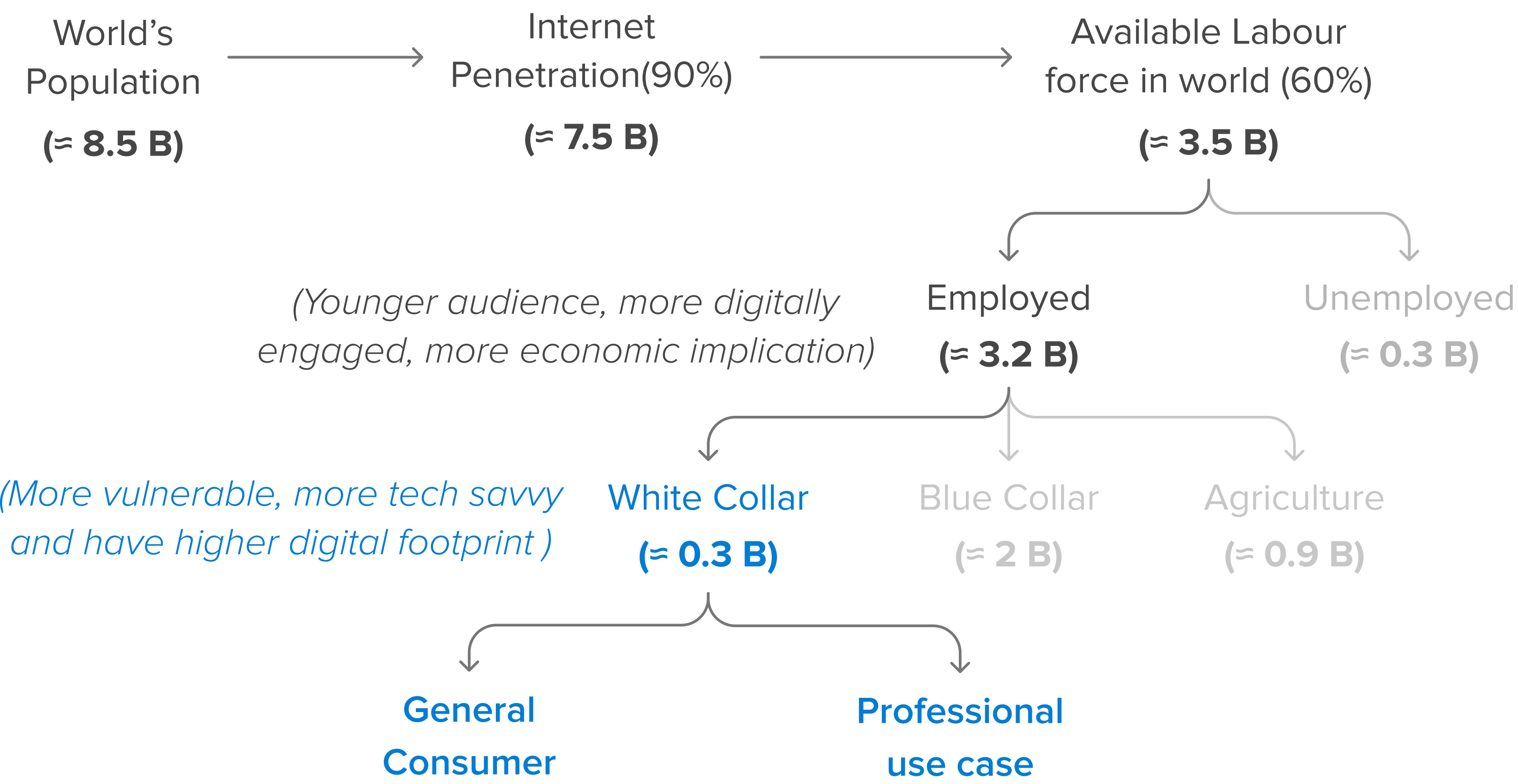
- Ethical Tech Development:** Companies will have to prioritize ethical, encrypted and privacy-centric design, reducing the negative externalities.
- Restore Trust and Adoption:** This can encourage greater adoption of new trustworthy online services, benefiting businesses and users alike.

Impact on Consumers

- Enhanced Individual Privacy:** People would have greater control over their data, reducing the risk of unauthorized data collection and misuse.
- Social and Psychological Well-being:** Enhanced privacy could contribute to improved mental well-being and reduced surveillance-related stress.

I will focus on consumer-side privacy to approach the problem from the demand side, which will ultimately foster trust and help businesses for long-term viability.

Who will you be solving for?



CRITERIA	GENERAL CONSUMER	PROFESSIONAL
Severity	High: The unauthorized access and deepfake attacks have a direct & immediate impact on personal safety, privacy, and psychological well-being.	Cushioned: Professionals may have organizational resources and existing security infrastructure to mitigate risks associated with data exploitation.
Scale of Impact	Broader: Solutions can cater to a diverse demographic given the widespread use of smart devices and social media.	Limited: Solutions may cater primarily to specific professional domains and organizational settings. Hence it narrows impact.
Market Opportunity	Higher: There will be a growing demand for privacy-centric solutions among general consumers as awareness of data privacy issues increases.	Competitive: Being a B2B model with professionals being stakeholders, it will require more resources and time to develop a competitive edge.

General Consumer



Rashi Jain

Tech Savvy Homeowner

Occupation: Graphic Designer

Location: Bangalore (Tier 1)

Technologies: Owns multiple smart home devices, smart car and is an active social media user

Pain Points

- Worries about data collected by her smart home devices being misused or accessed by unauthorized entities.
- Concerned about her digital footprint on social media leading to targeted deepfake attacks.
- Fears potential hacking of her smart car and home systems, threatening her safety and privacy.

Needs

- Wants to control the data collected by her smart home devices, tech companies and to opt-out of it.
- Seeks help to educate herself on maintaining digital privacy and security.

Professional



Shivendra Singh

Employee at a Tech company

Occupation: Product Manager

Location: Bangalore (Tier 1)

Technologies: Engages with advanced analytics tools, VR and relies heavily on his smartphone & other devices for work

Pain Points

- Anxious about corporate espionage through cyber-attacks leading to loss of sensitive project data.
- Worried about his professional network and interactions being exploited for fraudulent activities.
- Concerned about the implications of using predictive analytics and AI in product development

Needs

- Wants to train himself & his team on data handling & privacy when developing or using predictive analytics and AI.
- Requires measures to safeguard his professional network.

What part of the problem to solve first?

CRITERIA	REACH (10)	IMPACT (10)	CONFIDENCE (10)	EFFORT (5)	RICE SCORE (R*I*C/E)
Lack of education	8	6	7	2	168 ✓
Manage Personal encrypted Passwords & Documents	6	7	5	3	70
Monitor & Manage Data Shared Across Different Platforms	9	8	8	4	144 ✓
Precautions in Case of Data Breach	6	4	4	1	96
Deal with Harassment & Identity theft	5	8	5	5	40

Introducing Google Security

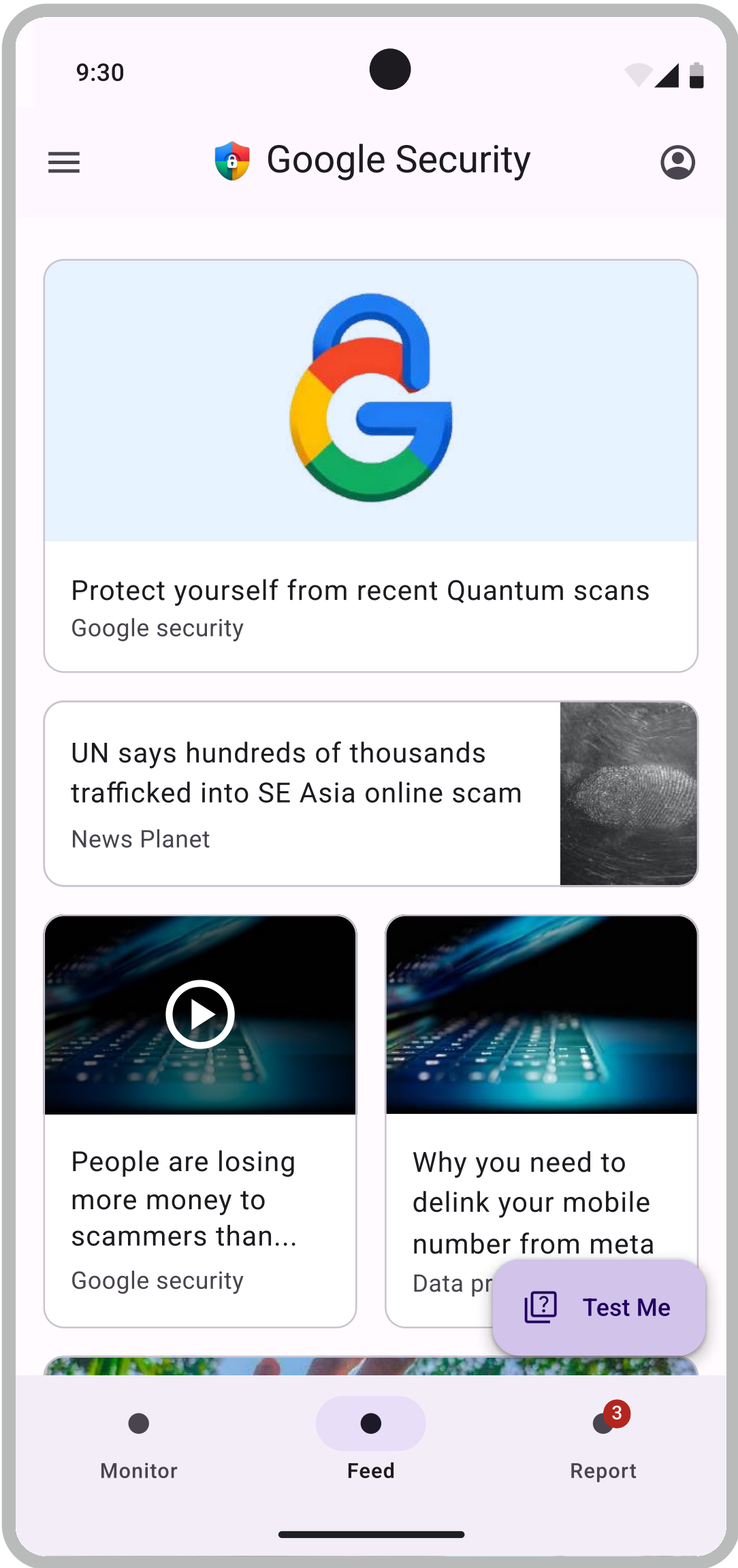
Solution approach

- The problem of digital footprints created using a service/product can be bifurcated into three parts, along with a solution at each step:
1. **Before Signing up for a service:** Educating the user about the privacy policies

2. **While using a service:** Assess what information can be prone to privacy breaches

3. **Service no longer required:** Revoke data access from organizations

1. Educating user about their data privacy



Feed section to educate users

Short videos & Articles

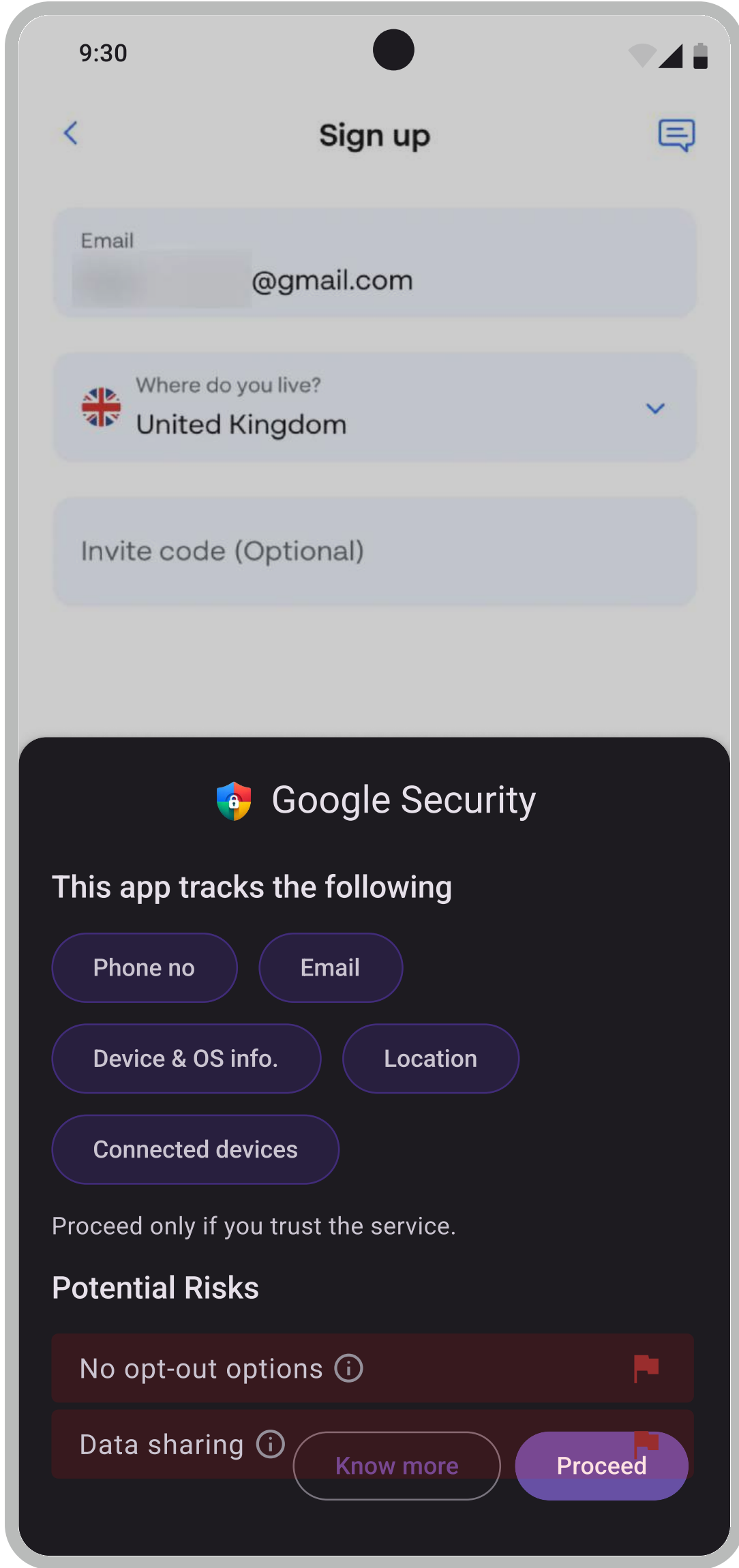
Dynamic feed that updates frequently to reflect the most recent threats and advice

Interactive Quiz

Gamified experience of real-world scenarios and common pitfalls related to digital privacy and scam prevention

Policy Summary

Summarizes the key points of privacy policies & cookies, highlighting potential red flags



Assessing Privacy Policy & Cookies

- Includes the short videos, articles and infographics

• Feed updates dynamically from time to time in accordance with the user’s data

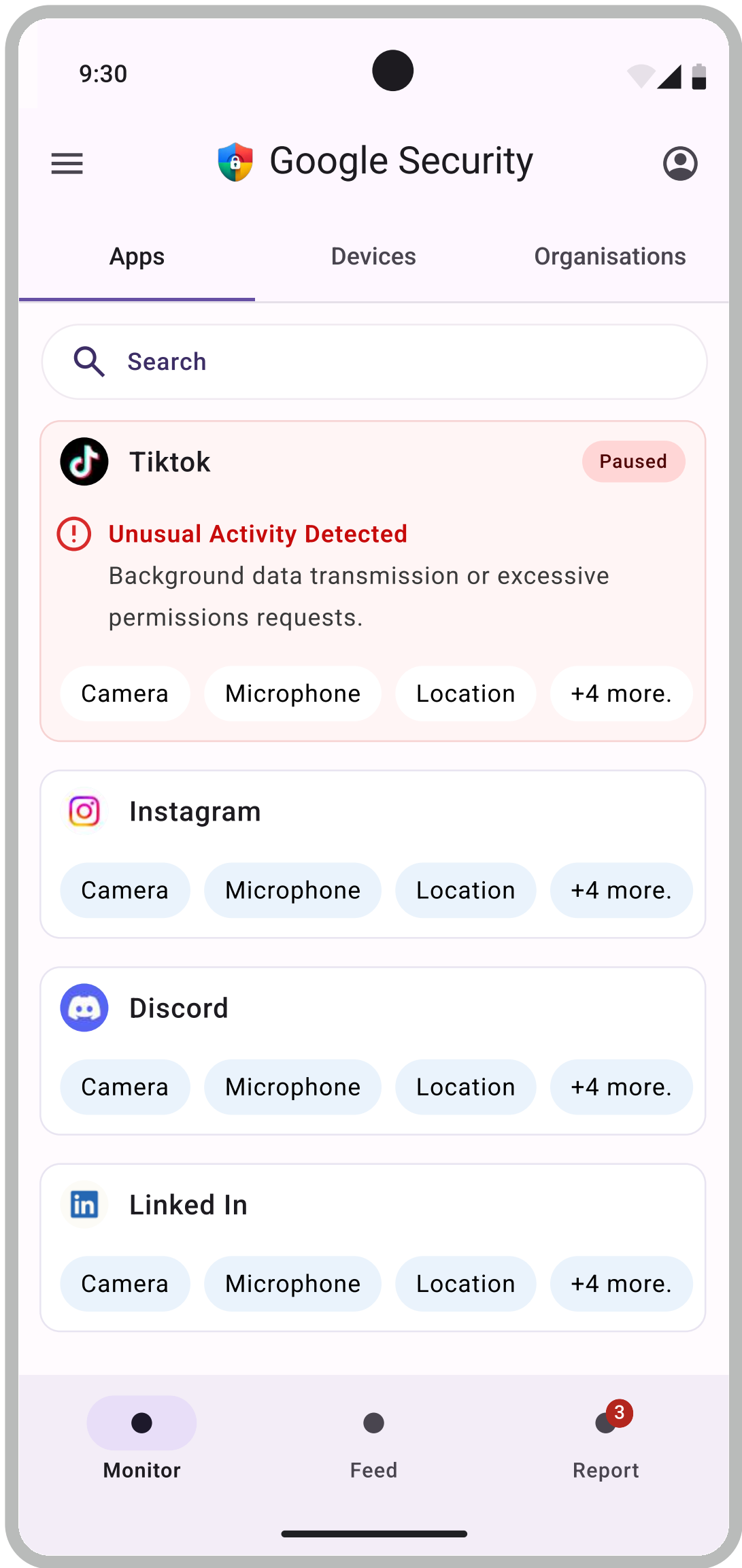
• Tailored interactive quizzes to teach users about do’s and don’ts while providing trending real-world pitfalls
- Activates when a user is about to sign up for a service/product

• Works with apps as well as browsers

• Provides privacy policy, cookies summary and potential red flags

• Can also have a browser extension for a multi-device experience

2. Track apps, devices and organisation holding data



List of installed & Uninstalled apps

Organizations

Data Brokers

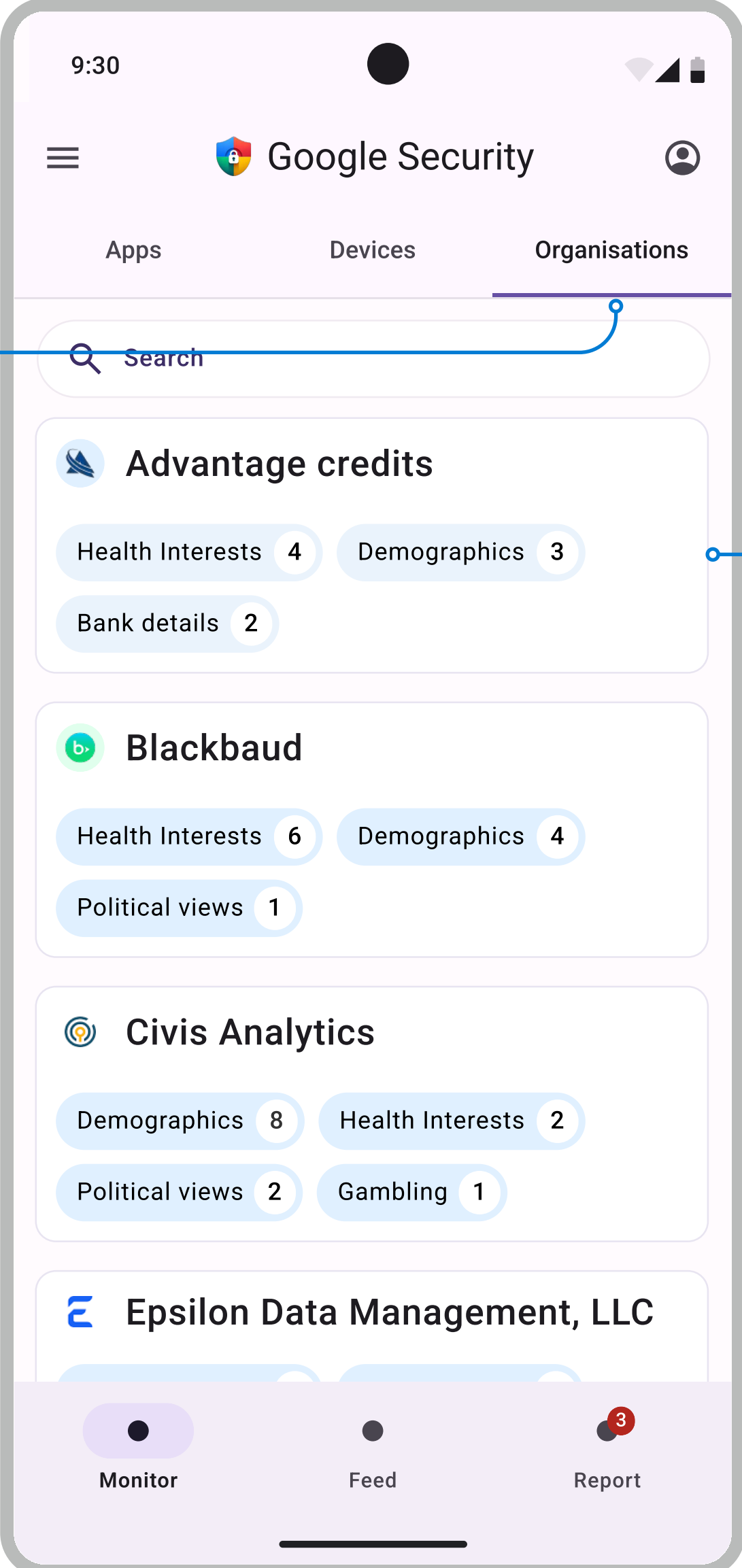
There are 230+ registered companies that create data profile of people across the world and further sells it to their client, which use it effectively for ads (e.g. Acxiom has data on 2.5 billion people with 10,000 attributes on each)

Past Associated Orgs

Includes academic and employment orgs

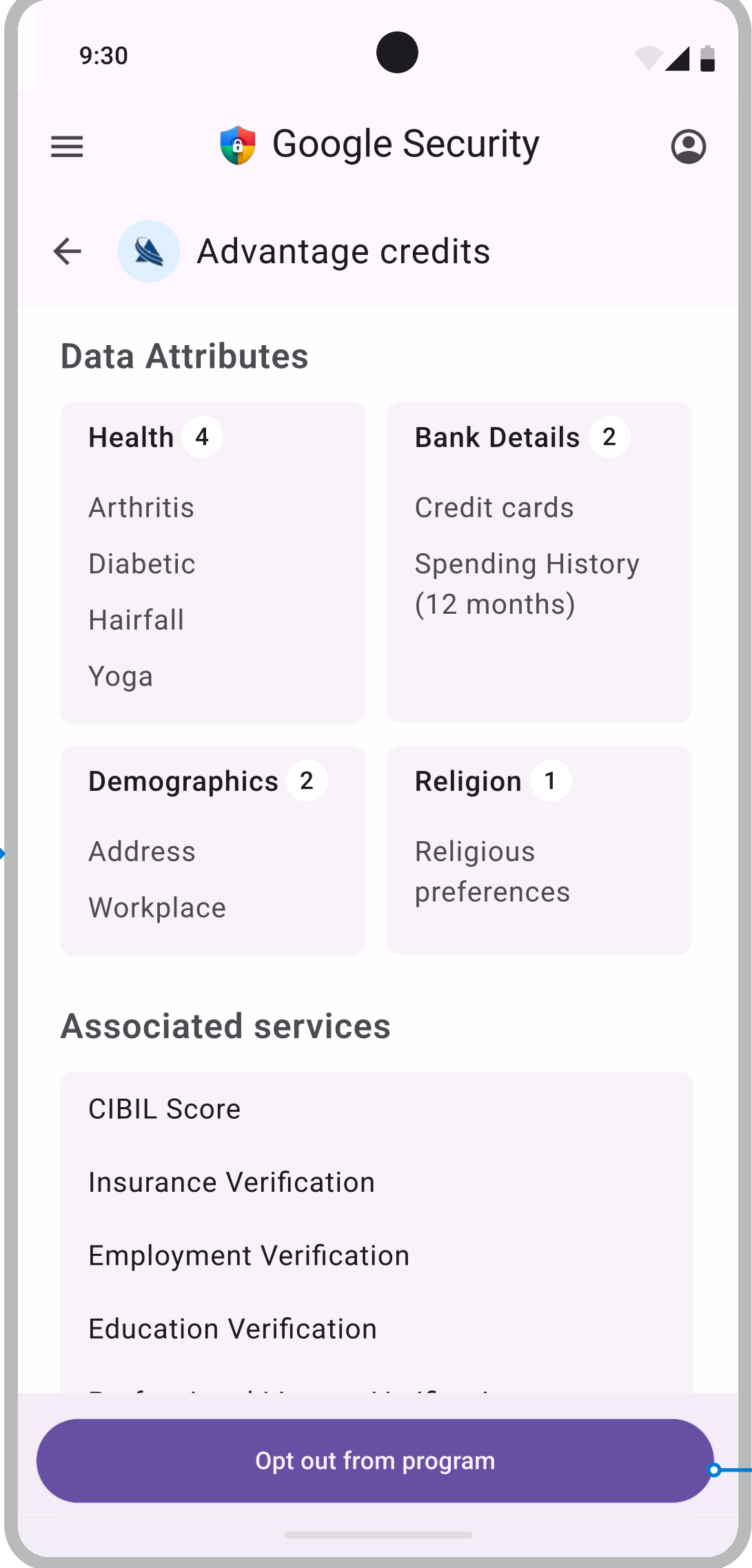
Government

Data stored by different government organizations

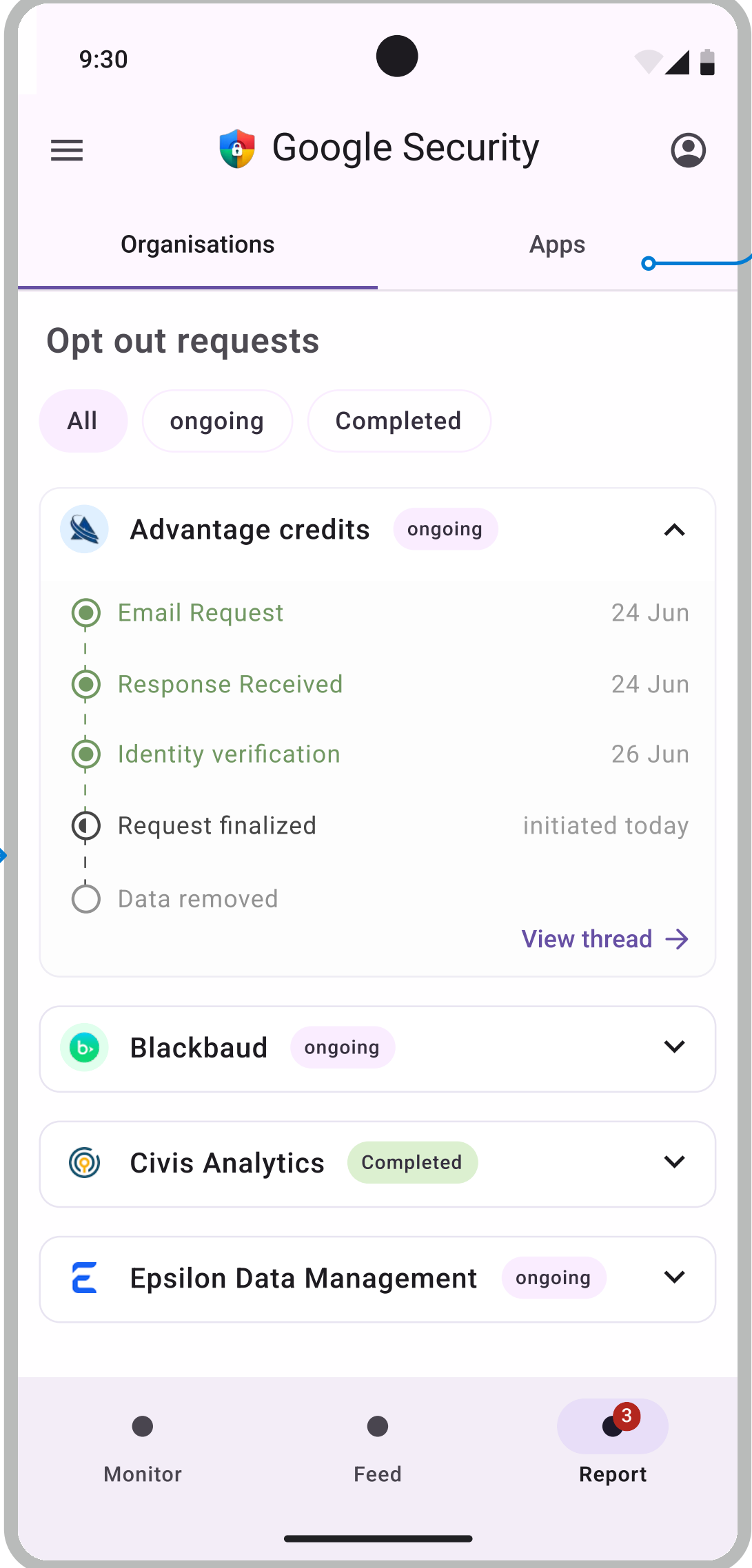


Organizations holding user data

3. Initiate data/account deletion request



Feed section to educate users



Feed section to educate users

Apps

In case of an app, user can initiate a automated account deletion process

Reports

The app would have access to BARD and capability to read/write emails, track & analyze progress, and complete tasks like identity verification.

Also, it can draft the legalities (according to country norms) in the case of any disparity.

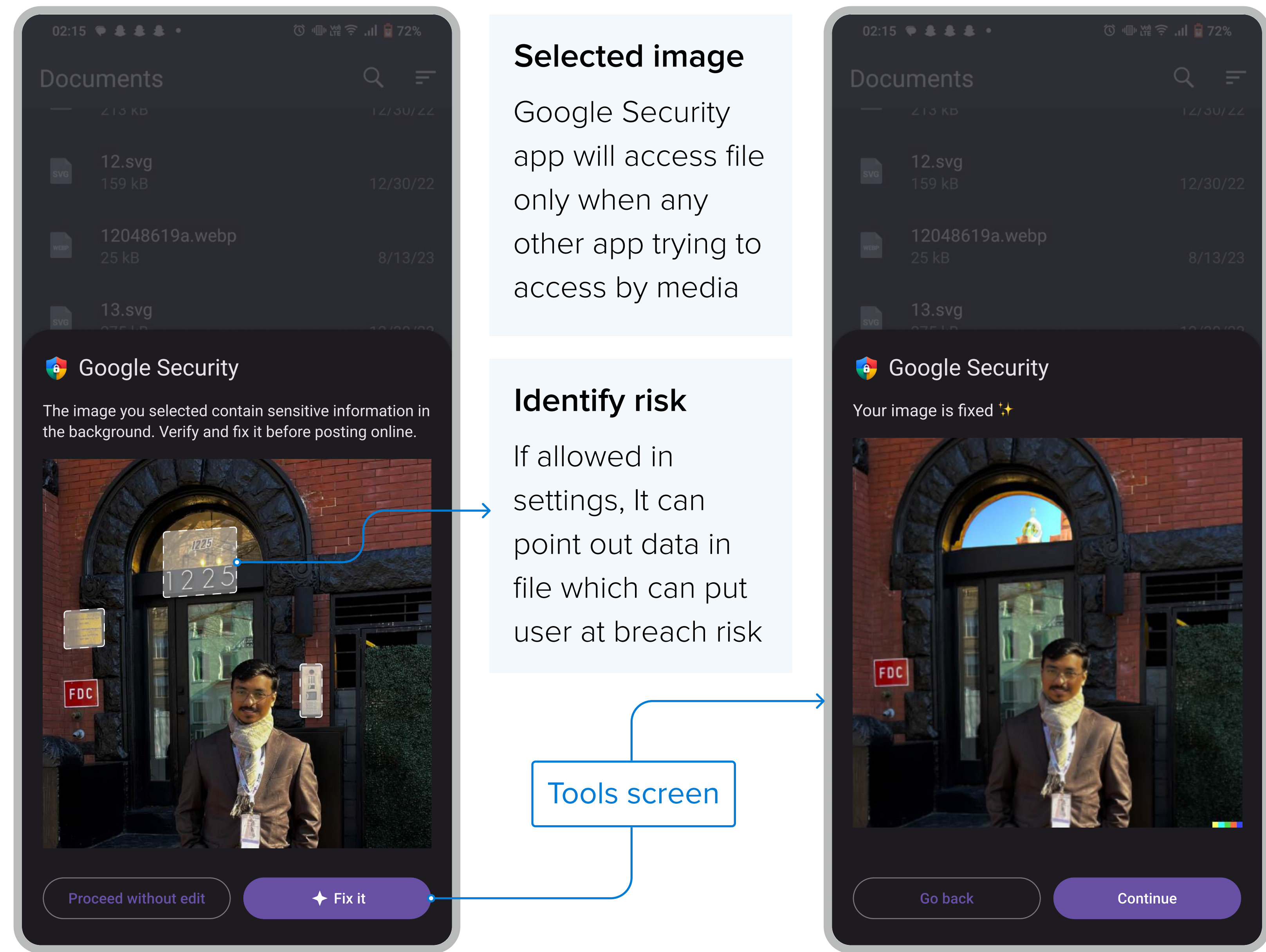
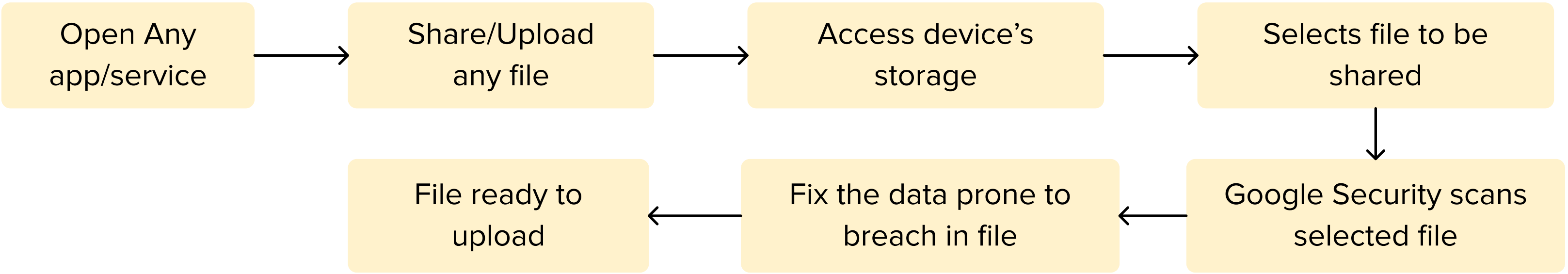
- Manage permissions and monitor unusual activity exhibited by apps & devices
 - Maintains history of transmitted data to servers and option to analyse log history
 - Option to initiate automatic account deletion request for uninstalled apps
 - Real time alerts in case of hack attempt
- List of all registered organizations which have data on user
 - A detailed list of all saved data attributes stored and associated services by each company (visible only to a user once the google account is verified with ID)
 - Option to Opt-Out (request to delete data)

- The user can initiate the Opt-out / Account deletion process, where the app automates all steps from writing request mail till the task completion
- Report section allows user to track the progress of opt out/account deletion task

Assumption: Government will mandate orgs to share data they are holding with respective user*

Note*: For now, only data broker organisations are mandated to register themselves and provide an "Opt out" option to people part of it's database. Also according to European countries, companies will be mandated to show what data they are holding in 2-3 years.

4. Assessing data about to be shared



- Capable of scanning the selected file (only when another app accesses local files) to identify any visible personal data that could potentially breach privacy norms (e.g., ticket PNR, financial details, ID numbers).

Feasibility of solutions

FEATURE	TECHNOLOGICAL FEASIBILITY	OPERATIONAL/REGULATORY/DEPENDENCY
Feed with videos & articles	Content-based Filtering to curate a personalized feed	Significant operational effort in update and maintenance of content
Interactive Quizzes	Adaptive Learning Algorithms to tailor quizzes based on the user's knowledge level. Also, we enable VR to make it more interactive	Will need to create a sandbox like environment for real life experience
Privacy Policy Summarizer	Utilize NLP to understand and summarize tedious privacy policies & cookies	Regulatory challenges may arise due to differing data protection laws across regions.
Unusual Activity Monitoring	Can use Neural Networks for auto-encoding of any anomaly detection	Users will have to trust our app to entrust with other app's monitoring
Data Transmission Monitoring	Can use DPI for transmission detection and NLP to parse and analyze log files	Robust infrastructure capable of handling a high volume of data
Automatic Account Deletion	Can use Google BARD APIs to automate the whole process	Will require access to Gmail and user's official documents
Scanning file for risk management	Can utilize OCR to read images and PDFs	Permission to access and edit files

What if I have to choose an MVP?

In case we have very limited resources available for the project, I would like to go with user education features. Because Google's news feed and rewards feature can provide an infrastructure for dynamic feed, hence reducing the efforts. Also, it is coupled with its extensive ecosystem and global trusted user base.

Success Metrics

North star Metric (User Adoption): **Average Privacy Score Improvement**

A metric that quantifies the extent to which users have been able to enhance their digital privacy using the features of the product.

$$\frac{[Sum\ of\ Post-Usage\ Privacy\ Scores] - [Sum\ of\ Pre-Usage\ Privacy\ Scores]}{Total\ Number\ of\ Users}$$

Metric #2 (Engagement): **Engagement Rate with Educational Content**

The frequency and depth of interactions (#visits, time spend, quiz attempted etc) indicates successful education and awareness creation among users regarding digital privacy issues.

$$\frac{Total\ Interaction\ with\ educational\ feed\ and\ quizzes}{Total\ Number\ of\ Users}$$

Metric #3 (Conversion): **Data Deletion/Opt-Out Requests Successful**

The number of successful data deletion or opt-out requests processed through the product's features. This reflects the direct impact on user privacy.

$$\frac{Total\ Number\ of\ Data\ Deletion/Opt-Out\ Requests\ completed\ successfully}{Total\ Number\ Requests\ Initiated}$$

Metric #4 (User Satisfaction): **Net Promoter Score**

Satisfaction scores would provide insight into how well the product meets user expectations and its effectiveness in addressing privacy concerns hence how likely are they to recommend the app to others.

How might this fail?

PITFALLS	MITIGATIONS
User’s Distrust in automated privacy features including other app history analyzation and account deletion process	• Transparent communication regarding how the features work and how data is handled. • Allow users to take control over automated features as much as possible.
Lack of engagement in educational material and quizzes	• Motivating users through gamification, leaderboards, badges and streaks • Inspirational stories on how the material helped other people
Inaccuracy in alerts by ML model during monitoring others apps and devices	• Constantly improve detection models and setup a feedback mechanism • Devising a system to let users adjust the setting in accordance to their specific needs and requirements
Unsuccessful data deletion requests due to incorporation from companies’ end	• Enable user to initiate process of a legal draft on their behalf

What else can be done?

We can empower our feed feature by creating a community platform around it to foster a collective sense of support and education among users around privacy issues. It can include Forums for discussion, a Privacy News Feed for staying updated on relevant issues, and Groups to offer & seek support for privacy-related issues & digital privacy challenges.